

Water supply and Sanitation

□ To impart knowledge to the students about the estimation of water demands

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 6361A is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 6361A.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Civil and Rural Engineering
Course code	6361A
Course title	Water supply and Sanitation
Semester	6 Credits: 4
Course category	Program Core
Periods per week	4 (L:4, T:0, P:0) Periods per semester:60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

17 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To impart knowledge to the students about the estimation of water demands for proper design of water supply works.
- To make the students aware about the different water conveyance and distribution systems.
- To Familiarize the students with different stages of water treatment, different methods of sewage treatment and disposal and operations involved.

Course outcomes

CO	Official outcome	Study level
CO1	determine water demands from available data for 14 Applying designing water supply systems.	See official syllabus
CO2	Suggest appropriate water treatment methods 15 Understanding Identify the components required for Building	See official syllabus
CO3	14 Understanding sanitation and Sewerage systems Suggest appropriate sewage treatment and disposal	See official syllabus
CO4	15 Understanding methods	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 3 3
CO2	CO2 3 3 3
CO3	CO3 3 3

Row	Extracted official mapping
CO4	CO4 3 3 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	demands from available data for designing water supply systems.	4	As specified
Module 2	Suggest appropriate water treatment methods	5	As specified
Module 3	Identify the components required for Building sanitation and Sewerage	4	As specified
Module 4	Suggest appropriate sewage treatment and disposal methods	4	As specified

MODULE 1

demands from available data for designing water supply systems.

This module is presented from the official syllabus scope for Water supply and Sanitation. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.

- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: demands from available data for designing water supply systems.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

3 Understanding sources of water. Determine water demands for a water supply

3 Understanding sources of water. Determine water demands for a water supply is an official topic in Module 1 of Water supply and Sanitation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding sources of water. Determine water demands for a water supply using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding sources of water. determine water demands for a water supply should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding sources of water. Determine water demands for a water supply means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Understanding sources of water. Determine water demands for a water supply ഓരോന്നിനും definition/meaning ഉണ്ടാകുന്നു. ഓരോന്നും ഓരോന്നും, steps, application ഉണ്ടാകുന്നു. Technical terms English-ൽ ഉണ്ടാകുന്നു.

4 Applying scheme Outline standard tests for analyzing water for

4 Applying scheme Outline standard tests for analyzing water for is an official topic in Module 1 of Water supply and Sanitation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying scheme Outline standard tests for analyzing water for using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying scheme outline standard tests for analyzing water for should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying scheme Outline standard tests for analyzing water for means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-4 Applying scheme Outline standard tests for analyzing water for definition/meaning .
 , steps, application . Technical terms English-

4 Understanding drinking purposes Explain the different rural water

4 Understanding drinking purposes Explain the different rural water is an official topic in Module 1 of Water supply and Sanitation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding drinking purposes Explain the different rural water using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding drinking purposes explain the different rural water should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding drinking purposes Explain the different rural water means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-4 Understanding drinking purposes Explain the different rural water definition/meaning definition/meaning. definition/meaning definition/meaning, steps, application definition/meaning definition/meaning. Technical terms English- definition/meaning definition/meaning.

3 Understanding supply schemes by the Government Sources of Water:

Surface source - lakes, streams, rivers and impounded reservoirs, Yield from surface source - Underground sources - springs, wells, infiltration wells and galleries - Yield from wells - test for yield. Intake Structures, Definition and types, Factors governing the location of an intake structure, Types of intakes. Demand of water: Factors affecting rate of demand, Variations of water demands, Forecasting of population, Methods of forecasting of population, (Simple problems on forecasting of population), Design period, Estimating of quantity of water supply required for city or town. Quality of water: Need for analysis of water, Characteristics of water - Physical, Chemical and Biological, Testing of water for Total solids, hardness, chlorides, dissolved Oxygen, pH, Fluoride, Nitrogen and its compounds, Bacteriological tests, E coli, B coli index, MPN, Sampling of water, Water quality standards as per IS 10500. Water supply schemes - Objectives, components, Rural water supply schemes by Government-NRDWP-Jal Jeevan Mission, Atal Bhujal Yojana, Jal Sakthi Abhiyan KRWSA-Jalanidhi, Jala Subhiksha.

3 Understanding supply schemes by the Government Sources of Water: Surface source - lakes, streams, rivers and impounded reservoirs, Yield from surface source - Underground sources - springs, wells, infiltration wells and galleries - Yield from wells - test for yield. Intake Structures, Definition and types, Factors governing the location of an intake structure, Types of intakes. Demand of water: Factors affecting rate of demand, Variations of water demands, Forecasting of population, Methods of forecasting of population, (Simple problems on forecasting of population), Design period, Estimating of quantity of water supply required for city or town. Quality of water: Need for analysis of water, Characteristics of water - Physical, Chemical and Biological, Testing of water for Total solids, hardness, chlorides, dissolved Oxygen, pH, Fluoride, Nitrogen and its compounds, Bacteriological tests, E coli, B coli index, MPN, Sampling of water, Water quality standards as per IS 10500. Water supply schemes - Objectives, components, Rural water supply schemes by Government-NRDWP-Jal Jeevan Mission, Atal Bhujal Yojana, Jal Sakthi Abhiyan KRWSA-Jalanidhi, Jala Subhiksha. is an official topic in Module 1 of Water supply and Sanitation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding supply schemes by the Government Sources of Water: Surface source - lakes, streams, rivers and impounded reservoirs, Yield from surface source - Underground sources - springs, wells, infiltration wells and galleries - Yield from wells - test for yield. Intake Structures, Definition and types, Factors governing the location of an intake structure, Types of intakes. Demand of water: Factors affecting rate of demand, Variations of water demands, Forecasting of population, Methods of forecasting of population, (Simple problems on forecasting of population), Design period, Estimating of quantity of water supply required for city or town. Quality of water: Need for analysis of water, Characteristics of water - Physical, Chemical and Biological, Testing of water for Total solids, hardness, chlorides, dissolved Oxygen, pH, Fluoride, Nitrogen and its compounds, Bacteriological tests, E coli, B coli index, MPN, Sampling of water, Water quality standards as per IS 10500. Water supply schemes - Objectives, components, Rural water supply schemes by Government-NRDWP-Jal Jeevan Mission, Atal Bhujal Yojana, Jal Sakthi Abhiyan KRWSA-Jalanidhi, Jala Subhiksha. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding supply schemes by the government sources of water: surface source - lakes, streams, rivers and impounded reservoirs, yield from surface source - underground sources - springs, wells, infiltration wells and galleries - yield from wells - test for yield. intake structures, definition and types, factors governing the location of an intake structure, types of intakes. demand of water: factors affecting rate of demand, variations of water demands, forecasting of population, methods of forecasting of population, (simple problems on forecasting of population), design period, estimating of quantity of water supply required for city or town. quality of water: need for analysis of water, characteristics of water - physical, chemical and biological, testing of water for total solids, hardness, chlorides, dissolved oxygen, ph, fluoride, nitrogen and its compounds, bacteriological tests, e coli, b coli index, mpn, sampling of water, water quality standards as per is 10500. water supply schemes - objectives, components, rural water supply schemes by government-nrdwp-jal jeevan mission, atal bhujal yojana, jal sakthi abhiyan krwsa-jalanidhi, jala subhiksha. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 3 Understanding supply schemes by the Government Sources of Water: Surface source - lakes, streams, rivers and impounded reservoirs, Yield from surface source - Underground sources - springs, wells, infiltration wells and galleries - Yield from wells - test for yield. Intake Structures, Definition and types, Factors governing the location of an intake structure, Types of intakes. Demand of water: Factors affecting rate of demand, Variations of water demands, Forecasting of population, Methods of forecasting of population, (Simple problems on forecasting of population), Design period, Estimating of quantity of water supply required for city or town. Quality of water: Need for analysis of water, Characteristics of water - Physical, Chemical and Biological, Testing of water for Total solids, hardness, chlorides, dissolved Oxygen, pH, Fluoride, Nitrogen and its compounds, Bacteriological tests, E coli, B coli index, MPN, Sampling of water, Water quality standards as per IS 10500. Water supply schemes - Objectives, components, Rural water supply schemes by Government-NRDWP-Jal Jeevan Mission, Atal Bhujal Yojana, Jal Sakthi Abhiyan KRWSA-Jalanidhi, Jala Subhiksha. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Understanding supply schemes by the Government Sources of Water: Surface source - lakes, streams, rivers and impounded reservoirs, Yield from surface source - Underground sources - springs, wells, infiltration wells and galleries - Yield from wells - test for yield. Intake Structures, Definition and types, Factors governing the location of an intake structure, Types of intakes. Demand of water: Factors affecting rate of demand, Variations of water demands, Forecasting of population, Methods of forecasting of population, (Simple problems on forecasting of population), Design period, Estimating of quantity of water supply required for city or town. Quality of water: Need for analysis of water, Characteristics of water - Physical, Chemical and Biological, Testing of water for Total solids, hardness, chlorides, dissolved Oxygen, pH, Fluoride, Nitrogen and its compounds, Bacteriological tests, E coli, B coli index, MPN, Sampling of water, Water quality standards as per IS 10500. Water supply schemes - Objectives, components, Rural water supply schemes by Government-NRDWP-Jal Jeevan Mission, Atal Bhujal Yojana, Jal Sakthi Abhiyan KRWSA-Jalanidhi, Jala Subhiksha. definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

demands from available data for designing water supply systems.

Describe various surface and sub surface

M1.01

3

Understanding

sources of water.

Determine water demands for a water supply

M1.02

4

Applying

scheme

Outline standard tests for analyzing water for

M1.03

4

Understanding

drinking purposes

Explain the different rural water

M1.04

3

Understanding

supply schemes by the Government

Contents:

Sources of Water: Surface source - lakes, streams, rivers and impounded reservoirs,

Yield from surface source - Underground sources - springs, wells, infiltration wells and

galleries - Yield from wells - test for yield. Intake Structures, Definition and types,

Factors governing the location of an intake structure, Types of intakes.

Demand of water: Factors affecting rate of demand, Variations of water demands,

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Suggest appropriate water treatment methods

This module is presented from the official syllabus scope for Water supply and Sanitation. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Suggest appropriate water treatment methods
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

sedimentation, and sedimentation with 3 Understanding coagulation. Outline the construction and operation of slow

sedimentation, and sedimentation with 3 Understanding coagulation. Outline the construction and operation of slow is an official topic in Module 2 of Water supply and Sanitation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify sedimentation, and sedimentation with 3 Understanding coagulation. Outline the construction and operation of slow using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, sedimentation, and sedimentation with 3 understanding coagulation. outline the construction and operation of slow should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What sedimentation, and sedimentation with 3 Understanding coagulation. Outline the construction and operation of slow means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഓക്സീജനേഷൻ, and sedimentation with 3 Understanding coagulation. Outline the construction and operation of slow sedimentation tank. ഓക്സീജനേഷൻ definition/meaning ഓക്സീജനേഷൻ. ഓക്സീജനേഷൻ steps, application ഓക്സീജനേഷൻ. Technical terms English-ഓക്സീജനേഷൻ.

3 Understanding sand, rapid sand and pressure filters. Explain methods of disinfection and

3 Understanding sand, rapid sand and pressure filters. Explain methods of disinfection and is an official topic in Module 2 of Water supply and Sanitation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding sand, rapid sand and pressure filters. Explain methods of disinfection and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding sand, rapid sand and pressure filters. explain methods of disinfection and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding sand, rapid sand and pressure filters. Explain methods of disinfection and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-3 Understanding sand, rapid sand and pressure filters. Explain methods of disinfection and ഓക്സീജനേഷൻ ഓക്സീജൻ definition/meaning ഓക്സീജനേഷൻ. ഓക്സീജൻ ഓക്സീജൻ ഓക്സീജൻ, steps, application ഓക്സീജൻ ഓക്സീജൻ. Technical terms English-ഓക്സീജൻ ഓക്സീജൻ.

advanced treatment methods for the water 3 Understanding in a given situation Identify the different components of a

advanced treatment methods for the water 3 Understanding in a given situation Identify the different components of a is an official topic in Module 2 of Water supply and Sanitation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify advanced treatment methods for the water 3 Understanding in a given situation Identify the different components of a using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, advanced treatment methods for the water 3 understanding in a given situation identify the different components of a should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What advanced treatment methods for the water 3 Understanding in a given situation Identify the different components of a means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പുറം advanced treatment methods for the water 3 Understanding in a given situation Identify the different components of a പദപരിചയം definition/meaning പദപരിചയം. പദപരിചയം പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms English-പദപരിചയം പദപരിചയം.

3 Understanding conveyance system Explain with sketches the methods of Applying

3 Understanding conveyance system Explain with sketches the methods of Applying is an official topic in Module 2 of Water supply and Sanitation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding conveyance system Explain with sketches the methods of Applying using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding conveyance system explain with sketches the methods of applying should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding conveyance system Explain with sketches the methods of Applying means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-3 Understanding conveyance system Explain with sketches the methods of Applying definition/meaning. steps, application. Technical terms English-

3 distribution of water in a given situation Purification of water: Objectives of water treatment, Aeration - objects and methods of aeration- Plain sedimentation, Sedimentation with coagulation, principles of coagulation, types of coagulants, Jar Test, process of coagulation, types of sedimentation tanks, Clariflocculator - Filtration -Mechanism of filtration- classification of filters: slow sand filter, rapid sand filter, pressure filter. Construction and working of slow sand and rapid sand filter, operational problems in filtration - Disinfection: Objects, methods of disinfection, Chlorination- Application of chlorine, forms of chlorination, types of chlorination practices, residual chlorine and its importance - Flow diagram of water treatment plants. (No design for treatment units). Introduction to water softening, iron removal and Deflouridation techniques-Nalgonda method. Advanced Water treatments- Electrolysis, Reverse Osmosis. Conveyance: Types of Pipes used for conveyance of water, choice of pipe material, Types of joints and valves - location and function on a pipe lane. Conservation of water- Rainwater Harvesting. Distribution of water: Methods of distribution of water- Gravity, pumping, and combined system, Service reservoirs - functions and types, Layouts of distribution of Water-Dead end system, grid iron system, circular system, radial system; their suitability, advantages and disadvantages. Hand pumps-types, installation, operation, and maintenance of hand pumps.

3 distribution of water in a given situation Purification of water: Objectives of water treatment, Aeration - objects and methods of aeration- Plain sedimentation, Sedimentation with coagulation, principles of coagulation, types of coagulants, Jar Test, process of coagulation, types of sedimentation tanks, Clariflocculator - Filtration -Mechanism of filtration- classification of filters: slow sand filter, rapid sand filter, pressure filter. Construction and working of slow sand and rapid sand filter, operational problems in filtration - Disinfection: Objects, methods of disinfection, Chlorination- Application of chlorine, forms of chlorination, types of chlorination practices, residual chlorine and its importance - Flow diagram of water treatment plants. (No design for treatment units). Introduction to water softening, iron removal and Deflouridation techniques- Nalgonda method. Advanced Water treatments-Electrolysis, Reverse Osmosis. Conveyance: Types of Pipes used for conveyance of water, choice of pipe material, Types of joints and valves - location and function on a pipe lane. Conservation of water- Rainwater Harvesting. Distribution of water: Methods of distribution of water- Gravity, pumping, and combined system, Service reservoirs - functions and types, Layouts of distribution of Water-Dead end system, grid iron system, circular system, radial system; their suitability, advantages and disadvantages. Hand pumps-types, installation, operation, and maintenance of hand pumps. is an official topic in Module 2 of Water supply and Sanitation. Study

the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 distribution of water in a given situation Purification of water: Objectives of water treatment, Aeration – objects and methods of aeration- Plain sedimentation, Sedimentation with coagulation, principles of coagulation, types of coagulants, Jar Test, process of coagulation, types of sedimentation tanks, Clariflocculator - Filtration -Mechanism of filtration- classification of filters: slow sand filter, rapid sand filter, pressure filter. Construction and working of slow sand and rapid sand filter, operational problems in filtration – Disinfection: Objects, methods of disinfection, Chlorination- Application of chlorine, forms of chlorination, types of chlorination practices, residual chlorine and its importance – Flow diagram of water treatment plants. (No design for treatment units). Introduction to water softening, iron removal and Defluoridation techniques-Nalgonda method. Advanced Water treatments-Electrolysis, Reverse Osmosis. Conveyance: Types of Pipes used for conveyance of water, choice of pipe material, Types of joints and valves - location and function on a pipe line. Conservation of water- Rainwater Harvesting. Distribution of water: Methods of distribution of water- Gravity, pumping, and combined system, Service reservoirs - functions and types, Layouts of distribution of Water-Dead end system, grid iron system, circular system, radial system; their suitability, advantages and disadvantages. Hand pumps-types, installation, operation, and maintenance of hand pumps. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 distribution of water in a given situation purification of water: objectives of water treatment, aeration – objects and methods of aeration- plain sedimentation, sedimentation with coagulation, principles of coagulation, types of coagulants, jar test, process of coagulation, types of sedimentation tanks, clariflocculator - filtration -mechanism of filtration- classification of filters: slow sand filter, rapid sand filter, pressure filter. construction and working of slow sand and rapid

sand filter, operational problems in filtration – disinfection: objects, methods of disinfection, chlorination- application of chlorine, forms of chlorination, types of chlorination practices, residual chlorine and its importance – flow diagram of water treatment plants. (no design for treatment units). introduction to water softening, iron removal and deflouridation techniques-nalgonda method. advanced water treatments- electrolysis, reverse osmosis. conveyance: types of pipes used for conveyance of water, choice of pipe material, types of joints and valves - location and function on a pipe lane. conservation of water- rainwater harvesting. distribution of water: methods of distribution of water- gravity, pumping, and combined system, service reservoirs - functions and types, layouts of distribution of water-dead end system, grid iron system, circular system, radial system; their suitability, advantages and disadvantages. hand pumps-types, installation, operation, and maintenance of hand pumps. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 distribution of water in a given situation Purification of water: Objectives of water treatment, Aeration – objects and methods of aeration- Plain sedimentation, Sedimentation with coagulation, principles of coagulation, types of coagulants, Jar Test, process of coagulation, types of sedimentation tanks, Clariflocculator - Filtration -Mechanism of filtration- classification of filters: slow sand filter, rapid sand filter, pressure filter. Construction and working of slow sand and rapid sand filter, operational problems in filtration – Disinfection: Objects, methods of disinfection, Chlorination- Application of chlorine, forms of chlorination, types of chlorination practices, residual chlorine and its importance – Flow diagram of water treatment plants. (No design for treatment units). Introduction to water softening, iron removal and Deflouridation techniques-Nalgonda method. Advanced Water treatments-Electrolysis, Reverse Osmosis. Conveyance: Types of Pipes used for conveyance of water, choice of pipe material, Types of joints and valves - location and function on a pipe lane. Conservation of water- Rainwater Harvesting. Distribution of water: Methods of distribution of water- Gravity, pumping, and combined system, Service reservoirs - functions and types, Layouts of distribution of Water-Dead end system, grid iron system, circular system, radial system; their suitability, advantages and disadvantages. Hand pumps-types, installation, operation, and maintenance of hand pumps. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 3 distribution of water in a given situation Purification of water: Objectives of water treatment, Aeration – objects and methods of aeration- Plain sedimentation, Sedimentation with coagulation,

principles of coagulation, types of coagulants, Jar Test, process of coagulation, types of sedimentation tanks, Clariflocculator - Filtration -Mechanism of filtration- classification of filters: slow sand filter, rapid sand filter, pressure filter. Construction and working of slow sand and rapid sand filter, operational problems in filtration - Disinfection: Objects, methods of disinfection, Chlorination- Application of chlorine, forms of chlorination, types of chlorination practices, residual chlorine and its importance - Flow diagram of water treatment plants. (No design for treatment units). Introduction to water softening, iron removal and Deflouridation techniques-Nalgonda method. Advanced Water treatments- Electrolysis, Reverse Osmosis. Conveyance: Types of Pipes used for conveyance of water, choice of pipe material, Types of joints and valves - location and function on a pipe line. Conservation of water- Rainwater Harvesting. Distribution of water: Methods of distribution of water- Gravity, pumping, and combined system, Service reservoirs - functions and types, Layouts of distribution of Water-Dead end system, grid iron system, circular system, radial system; their suitability, advantages and disadvantages. Hand pumps-types, installation, operation, and maintenance of hand pumps. **ಪ್ರಮುಖ ಪದಗಳ ಅರ್ಥ/ಮಾನಾ** **ಪ್ರಮುಖ ಪದಗಳ ಅರ್ಥ/ಮಾನಾ**. **ಪ್ರಮುಖ ಪದಗಳ ಅರ್ಥ/ಮಾನಾ**, steps, application **ಪ್ರಮುಖ ಪದಗಳ ಅರ್ಥ/ಮಾನಾ** **ಪ್ರಮುಖ ಪದಗಳ ಅರ್ಥ/ಮಾನಾ**. Technical terms English-**ಪ್ರಮುಖ ಪದಗಳ ಅರ್ಥ/ಮಾನಾ**.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Suggest appropriate water treatment methods

M2.01 Explain methods of aeration, plain sedimentation, and sedimentation with coagulation. 3

Understanding Outline the construction and operation of slow sand, rapid sand and pressure filters. 3

M2.02 Explain methods of disinfection and advanced treatment methods for the water in a given situation 3

Understanding Identify the different components of a conveyance system 3

Applying Explain with sketches the methods of distribution of water in a given situation 3

M2.05 Series Test - I 1

Contents:

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Identify the components required for Building sanitation and Sewerage

This module is presented from the official syllabus scope for Water supply and Sanitation. Use the topic cards for learning and the source transcript for exact

coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Identify the components required for Building sanitation and Sewerage
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

3 Remembering sullage, and garbage

3 Remembering sullage, and garbage is an official topic in Module 3 of Water supply and Sanitation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Remembering sullage, and garbage using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 remembering sullage, and garbage should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Remembering sullage, and garbage means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-3 Remembering sullage, and garbage
സുല്ല്യജ്വലനം definition/meaning സുല്ല്യജ്വലനം. സുല്ല്യജ്വലനം
steps, application സുല്ല്യജ്വലനം. Technical terms English-സുല്ല്യജ്വലനം
സുല്ല്യജ്വലനം.

Identify the different building sanitary fittings 4 Understanding and systems of plumbing Explain different sewerage

Identify the different building sanitary fittings 4 Understanding and systems of plumbing Explain different sewerage is an official topic in Module 3 of Water supply and Sanitation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify the different building sanitary fittings 4 Understanding and systems of plumbing Explain different sewerage using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify the different building sanitary fittings 4 understanding and systems of plumbing explain different sewerage should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify the different building sanitary fittings 4 Understanding and systems of plumbing Explain different sewerage means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓക്സ്ജനേഷൻ Identify the different building sanitary fittings 4 Understanding and systems of plumbing Explain different sewerage ഓക്സ്ജനേഷൻ ഓക്സ്ജനേഷൻ definition/meaning ഓക്സ്ജനേഷൻ. ഓക്സ്ജനേഷൻ ഓക്സ്ജനേഷൻ steps, application ഓക്സ്ജനേഷൻ ഓക്സ്ജനേഷൻ. Technical terms English-ഓക്സ്ജനേഷൻ ഓക്സ്ജനേഷൻ.

systems

systems is an official topic in Module 3 of Water supply and Sanitation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify systems using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, systems should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What systems means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-എ സിസ്റ്റംസ് നിർവ്വചനം/അർത്ഥം നിർവ്വചിക്കുക. നിർവ്വചനം നിർവ്വചിക്കുക, steps, application നിർവ്വചിക്കുക നിർവ്വചിക്കുക. Technical terms English-എ നിർവ്വചിക്കുക നിർവ്വചിക്കുക.

Describe the various types of sewer 3 Understanding appurtenances Building Sanitation: Necessity of sanitation, Necessity to treat domestic sewage, Definitions - Sewage, sullage, types of sewage. Definition of the terms related to Building Sanitation- Water pipe, Rain water pipe, Soil pipe, Sullage pipe, Vent pipe. Building Sanitary fittings- Water closet - Indian and European type, flushing cistern, wash basin, sinks, Urinals. Traps-types, qualities of good trap. Systems of plumbing - one pipe, two pipe, single stack, choice of system. Principles regarding design of building drainage, inspection and junction chambers, their necessity, location, size and shape. Systems of Sewerage and Sewer Appurtenances: Types of Sewers, Systems of sewerage, self-cleansing velocity and non-scouring velocity, Laying, Testing and maintenance of sewers, Manholes and Drop Manhole-component parts, location, spacing, construction details, Sewer Inlets, Street Inlets.

Describe the various types of sewer 3 Understanding appurtenances Building Sanitation: Necessity of sanitation, Necessity to treat domestic sewage, Definitions - Sewage, sullage, types of sewage. Definition of the terms related to Building Sanitation- Water pipe, Rain water pipe, Soil pipe, Sullage pipe, Vent pipe. Building Sanitary fittings- Water closet - Indian and European type, flushing cistern, wash basin, sinks, Urinals. Traps-types, qualities of good trap. Systems of plumbing - one pipe, two pipe, single stack, choice of system. Principles regarding design of building drainage, inspection and junction chambers, their necessity, location, size and shape. Systems of Sewerage and Sewer Appurtenances: Types of Sewers, Systems of sewerage, self-cleansing velocity and non-scouring velocity, Laying, Testing and maintenance of sewers, Manholes and Drop Manhole-component parts, location, spacing, construction details, Sewer Inlets, Street Inlets. is an official topic in Module 3 of Water supply and Sanitation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the various types of sewer 3 Understanding appurtenances Building Sanitation: Necessity of sanitation, Necessity to treat domestic sewage, Definitions - Sewage, sullage, types of sewage. Definition of the terms related to Building Sanitation- Water pipe, Rain water pipe, Soil pipe, Sullage

pipe, Vent pipe. Building Sanitary fittings- Water closet – Indian and European type, flushing cistern, wash basin, sinks, Urinals. Traps-types, qualities of good trap. Systems of plumbing - one pipe, two pipe, single stack, choice of system. Principles regarding design of building drainage, inspection and junction chambers, their necessity, location, size and shape. Systems of Sewerage and Sewer Appurtenances: Types of Sewers, Systems of sewerage, self-cleansing velocity and non-scouring velocity, Laying, Testing and maintenance of sewers, Manholes and Drop Manhole-component parts, location, spacing, construction details, Sewer Inlets, Street Inlets. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the various types of sewer 3 understanding appurtenances building sanitation: necessity of sanitation, necessity to treat domestic sewage, definitions - sewage, sullage, types of sewage. definition of the terms related to building sanitation- water pipe, rain water pipe, soil pipe, sullage pipe, vent pipe. building sanitary fittings- water closet – indian and european type, flushing cistern, wash basin, sinks, urinals. traps-types, qualities of good trap. systems of plumbing - one pipe, two pipe, single stack, choice of system. principles regarding design of building drainage, inspection and junction chambers, their necessity, location, size and shape. systems of sewerage and sewer appurtenances: types of sewers, systems of sewerage, self-cleansing velocity and non-scouring velocity, laying, testing and maintenance of sewers, manholes and drop manhole-component parts, location, spacing, construction details, sewer inlets, street inlets. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the various types of sewer 3 Understanding appurtenances Building Sanitation: Necessity of sanitation, Necessity to treat domestic sewage, Definitions - Sewage, sullage, types of sewage. Definition of the terms related to Building Sanitation- Water pipe, Rain water pipe, Soil pipe, Sullage pipe, Vent pipe. Building Sanitary fittings- Water closet – Indian and European type, flushing cistern, wash basin, sinks, Urinals. Traps-types, qualities of good trap. Systems of plumbing - one pipe, two pipe, single stack, choice of system. Principles regarding design of building drainage, inspection and junction chambers, their necessity, location, size and shape. Systems of Sewerage and Sewer Appurtenances: Types of Sewers, Systems of sewerage, self-cleansing velocity and non-scouring velocity, Laying, Testing and maintenance of sewers, Manholes and Drop Manhole-component parts, location, spacing, construction details, Sewer Inlets, Street Inlets. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Describe the various types of sewer 3 Understanding appurtenances Building Sanitation: Necessity of sanitation, Necessity to treat domestic sewage, Definitions - Sewage, sullage, types of sewage. Definition of the terms related to Building Sanitation- Water pipe, Rain water pipe, Soil pipe, Sullage pipe, Vent pipe. Building Sanitary fittings- Water closet – Indian and European type, flushing cistern, wash basin, sinks, Urinals. Traps-types, qualities of good trap. Systems of plumbing - one pipe, two pipe, single stack, choice of system. Principles regarding design of building drainage, inspection and junction chambers, their necessity, location, size and shape. Systems of Sewerage and Sewer Appurtenances: Types of Sewers, Systems of sewerage, self-cleansing velocity and non-scouring velocity, Laying, Testing and maintenance of sewers, Manholes and Drop Manhole-component parts, location, spacing, construction details, Sewer Inlets, Street Inlets. **പ്രധാനപ്പെട്ട പദങ്ങളുടെ definition/meaning** **പ്രധാനപ്പെട്ട പദങ്ങളുടെ** **പ്രധാനപ്പെട്ട പദങ്ങളുടെ**, steps, application **പ്രധാനപ്പെട്ട പദങ്ങളുടെ** **പ്രധാനപ്പെട്ട പദങ്ങളുടെ**. Technical terms English- **പ്രധാനപ്പെട്ട പദങ്ങളുടെ**.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Identify the components required for Building sanitation and Sewerage systems

M3.01	Define Sewage, Sewerage, Sewer, refuse, sullage, and garbage	3
Remembering		

M3.02	Identify the different building sanitary fittings and systems of plumbing	4
Understanding		

M3.03	Explain different sewerage systems	4
Understanding		

M3.04	Describe the various types of sewer appurtenances	3
Understanding		

Contents:

Building Sanitation: Necessity of sanitation, Necessity to treat domestic sewage, Definitions - Sewage, sullage, types of sewage. Definition of the terms related to Building

Sanitation- Water pipe, Rain water pipe, Soil pipe, Sullage pipe, Vent pipe. Building

Sanitary fittings- Water closet – Indian and European type, flushing cistern, wash basin, sinks, Urinals. Traps-types, qualities of good trap. Systems of plumbing - one pipe, two

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Suggest appropriate sewage treatment and disposal methods

This module is presented from the official syllabus scope for Water supply and Sanitation. Use the topic cards for learning and the source transcript for exact

coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Suggest appropriate sewage treatment and disposal methods
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

3 Understanding Outline the functions and working of screens,

3 Understanding Outline the functions and working of screens, is an official topic in Module 4 of Water supply and Sanitation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Outline the functions and working of screens, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding outline the functions and working of screens, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding Outline the functions and working of screens, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-3 Understanding Outline the functions and working of screens, definition/meaning. steps, application. Technical terms English-

3 Understanding skimming tanks, and grit chambers. Describe various sewage treatment and disposal

3 Understanding skimming tanks, and grit chambers. Describe various sewage treatment and disposal is an official topic in Module 4 of Water supply and Sanitation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding skimming tanks, and grit chambers. Describe various sewage treatment and disposal using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding skimming tanks, and grit chambers. describe various sewage treatment and disposal should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding skimming tanks, and grit chambers. Describe various sewage treatment and disposal means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-3 Understanding skimming tanks, and grit chambers. Describe various sewage treatment and disposal ഓരോന്നിന്റെയും definition/meaning. ഓരോന്നിന്റെയും steps, application ഓരോന്നിന്റെയും Technical terms English-ൽ പറയുക.

4 Understanding methods Explain various low cost rural sanitation

4 Understanding methods Explain various low cost rural sanitation is an official topic in Module 4 of Water supply and Sanitation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding methods Explain various low cost rural sanitation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding methods explain various low cost rural sanitation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding methods Explain various low cost rural sanitation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4- 4 Understanding methods Explain various low cost rural sanitation ഓരോന്നിന്റെയും definition/meaning എഴുതുക. ഓരോന്നിന്റെയും steps, application എഴുതുക. Technical terms English-ൽ എഴുതുക.

5 Understanding systems Analysis of sewage: Characteristics of sewage, B.O.D., C.O.D. and its significance, Central Pollution Control Board Norms for discharge of treated sewage, Objects of sewage treatment and flow diagram of conventional sewage treatment plant. Treatment of Sewage: Screening, Types of screens, Grit removal, Skimming, Sedimentation of sewage, Aerobic and anaerobic process, Sludge digestion, trickling filters, Activated sludge process, Disposal of sewage, Oxidation Pond, Oxidation ditch, aerated lagoons, anaerobic lagoon. Rural sanitation: Low cost rural sanitation systems-Privies of a conservancy system of sanitation- Bore hole privy, Aqua privy, Sulabh Shauchalya, Two pit pour flush water seal, Septic tank and soak pit, Recycling and Reuse of domestic waste. Composting (Nadep or Vermiculture), Biogas plants. Low cost community and individual Garbage disposal systems, Ferro-cement storage tanks. Rural sanitation programmes by Government of India, Swachh Bharat Mission-Objectives and strategies. Suchitwa mission. Text / Reference: T/R Book Title/Author S.C. Rangwala: Water Supply and Sanitary Engineering; Charotar Publishing T1 House R1 Garg, S.K., Environmental Engineering Vol. I and Vol. II, Khanna Publishers R2 K.N.Duggal : Elements of Public Health Engineering ; S.Chand & Co. R3 Sharma S.C, Environmental Engineering, Khanna Publishing House, New Delhi2 R4 Birdie, G. S. and Birdie, J.S.Water Supply and Sanitary Engineering, Dhanpat Rai R5 Desai, Vasant, Rural Development in India: Past, Present and Future: a challenge in the crisis, Himalaya Publishing House, Delhi. R6 Singh, Katar, Rural Development Principles, Policies and Management, Sage Publications India Pvt Ltd. Online Resources: Sl.No Website Link 1 <http://nptel.ac.in/courses> 2 <http://www.vlab.co.in/ba-nptel-labs-civil-engineering> 3 Document Published by Ministry of Rural development, Govt. of India, Ministry of Rural development.

5 Understanding systems Analysis of sewage: Characteristics of sewage, B.O.D., C.O.D. and its significance, Central Pollution Control Board Norms for discharge of treated sewage, Objects of sewage treatment and flow diagram of conventional sewage treatment plant. Treatment of Sewage: Screening, Types of screens, Grit removal, Skimming, Sedimentation of sewage, Aerobic and anaerobic process, Sludge digestion, trickling filters, Activated sludge process, Disposal of sewage, Oxidation Pond, Oxidation ditch, aerated lagoons, anaerobic lagoon. Rural sanitation: Low cost rural sanitation systems-Privies of a conservancy system of sanitation- Bore hole privy, Aqua privy, Sulabh Shauchalya, Two pit pour flush water seal, Septic tank and soak pit, Recycling and Reuse of domestic waste. Composting (Nadep or Vermiculture), Biogas plants. Low cost community and individual Garbage disposal systems, Ferro-cement storage tanks. Rural sanitation programmes by Government of India, Swachh Bharat Mission-

Objectives and strategies. Suchitwa mission. Text / Reference: T/R Book Title/Author S.C. Rangwala: Water Supply and Sanitary Engineering; Charotar Publishing T1 House R1 Garg, S.K., Environmental Engineering Vol. I and Vol. II, Khanna Publishers R2 K.N.Duggal : Elements of Public Health Engineering ; S.Chand & Co. R3 Sharma S.C, Environmental Engineering, Khanna Publishing House, New Delhi2 R4 Birdie, G. S. and Birdie, J.S.Water Supply and Sanitary Engineering, Dhanpat Rai R5 Desai, Vasant, Rural Development in India: Past, Present and Future: a challenge in the crisis, Himalaya Publishing House, Delhi. R6 Singh, Katar, Rural Development Principles, Policies and Management, Sage Publications India Pvt Ltd. Online Resources: Sl.No Website Link 1 <http://nptel.ac.in/courses> 2 <http://www.vlab.co.in/ba-nptel-labs-civil-engineering> 3 Document Published by Ministry of Rural development, Govt. of India, Ministry of Rural development. is an official topic in Module 4 of Water supply and Sanitation. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding systems Analysis of sewage: Characteristics of sewage, B.O.D., C.O.D. and its significance, Central Pollution Control Board Norms for discharge of treated sewage, Objects of sewage treatment and flow diagram of conventional sewage treatment plant. Treatment of Sewage: Screening, Types of screens, Grit removal, Skimming, Sedimentation of sewage, Aerobic and anaerobic process, Sludge digestion, trickling filters, Activated sludge process, Disposal of sewage, Oxidation Pond, Oxidation ditch, aerated lagoons, anaerobic lagoon. Rural sanitation: Low cost rural sanitation systems-Privies of a conservancy system of sanitation- Bore hole privy, Aqua privy, Sulabh Shauchalya, Two pit pour flush water seal, Septic tank and soak pit, Recycling and Reuse of domestic waste. Composting (Nadep or Vermiculture), Biogas plants. Low cost community and individual Garbage disposal systems, Ferro-cement storage tanks. Rural sanitation programmes by Government of India, Swachh Bharat Mission-Objectives and strategies. Suchitwa mission. Text / Reference: T/R Book Title/Author S.C. Rangwala: Water Supply and Sanitary Engineering; Charotar Publishing T1 House R1 Garg, S.K., Environmental Engineering Vol. I and Vol. II, Khanna Publishers R2 K.N.Duggal : Elements of Public Health Engineering ; S.Chand & Co. R3 Sharma S.C, Environmental Engineering, Khanna Publishing House, New Delhi2 R4 Birdie, G. S. and Birdie, J.S.Water Supply and Sanitary Engineering, Dhanpat Rai R5 Desai,

Vasant, Rural Development in India: Past, Present and Future: a challenge in the crisis, Himalaya Publishing House, Delhi. R6 Singh, Katar, Rural Development Principles, Policies and Management, Sage Publications India Pvt Ltd. Online Resources: Sl.No Website Link 1 <http://nptel.ac.in/courses> 2 <http://www.vlab.co.in/ba-nptel-labs-civil-engineering> 3 Document Published by Ministry of Rural development, Govt. of India, Ministry of Rural development. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding systems analysis of sewage: characteristics of sewage, b.o.d., c.o.d. and its significance, central pollution control board norms for discharge of treated sewage, objects of sewage treatment and flow diagram of conventional sewage treatment plant. treatment of sewage: screening, types of screens, grit removal, skimming, sedimentation of sewage, aerobic and anaerobic process, sludge digestion, trickling filters, activated sludge process, disposal of sewage, oxidation pond, oxidation ditch, aerated lagoons, anaerobic lagoon. rural sanitation: low cost rural sanitation systems-privies of a conservancy system of sanitation- bore hole privy, aqua privy, sulabh shauchalya, two pit pour flush water seal, septic tank and soak pit, recycling and reuse of domestic waste. composting (nadep or vermiculture), biogas plants. low cost community and individual garbage disposal systems, ferro-cement storage tanks. rural sanitation programmes by government of india, swachh bharat mission-objectives and strategies. suchitwa mission. text / reference: t/r book title/author s.c. rangwala: water supply and sanitary engineering; charotar publishing t1 house r1 garg, s.k., environmental engineering vol. i and vol. ii, khanna publishers r2 k.n.duggal : elements of public health engineering ; s.chand & co. r3 sharma s.c, environmental engineering, khanna publishing house, new delhi2 r4 birdie, g. s. and birdie, j.s.water supply and sanitary engineering, dhanpat rai r5 desai, vasant, rural development in india: past, present and future: a challenge in the crisis, himalaya publishing house, delhi. r6 singh, katar, rural development principles, policies and management, sage publications india pvt ltd. online resources: sl.no website link 1 <http://nptel.ac.in/courses> 2 <http://www.vlab.co.in/ba-nptel-labs-civil-engineering> 3 document published by ministry of rural development, govt. of india, ministry of rural development. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 5 Understanding systems Analysis of sewage: Characteristics of sewage, B.O.D., C.O.D. and its significance, Central Pollution Control Board Norms for discharge of treated sewage, Objects of sewage treatment and flow diagram of conventional sewage treatment plant. Treatment of Sewage: Screening, Types of screens, Grit removal, Skimming, Sedimentation of sewage, Aerobic and anaerobic process, Sludge digestion, trickling filters, Activated sludge process, Disposal of sewage, Oxidation Pond, Oxidation ditch, aerated lagoons, anaerobic lagoon. Rural sanitation: Low cost rural sanitation systems-Privies of a conservancy system of sanitation- Bore hole privy, Aqua privy, Sulabh Shauchalya, Two pit pour flush water seal, Septic tank and soak pit, Recycling and Reuse of domestic waste. Composting (Nadep or Vermiculture), Biogas plants. Low cost community and individual Garbage disposal systems, Ferro-cement storage tanks. Rural sanitation programmes by Government of India, Swachh Bharat Mission-Objectives and strategies. Suchitwa mission. Text / Reference: T/R Book Title/Author S.C. Rangwala: Water Supply and Sanitary Engineering; Charotar Publishing T1 House R1 Garg, S.K., Environmental Engineering Vol. I and Vol. II, Khanna Publishers R2 K.N.Duggal : Elements of Public Health Engineering ; S.Chand & Co. R3 Sharma S.C, Environmental Engineering, Khanna Publishing House, New Delhi R4 Birdie, G. S. and Birdie, J.S.Water Supply and Sanitary Engineering, Dhanpat Rai R5 Desai, Vasant, Rural Development in India: Past, Present and Future: a challenge in the crisis, Himalaya Publishing House, Delhi. R6 Singh, Katar, Rural Development Principles, Policies and Management, Sage Publications India Pvt Ltd. Online Resources: Sl.No Website Link 1 http://nptel.ac.in/courses/2 http://www.vlab.co.in/ba-nptel-labs-civil-engineering 3 Document Published by Ministry of Rural development, Govt. of India, Ministry of Rural development. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-5 Understanding systems Analysis of sewage: Characteristics of sewage, B.O.D., C.O.D. and its significance, Central Pollution Control Board Norms for discharge of treated sewage, Objects of sewage treatment and flow diagram of conventional sewage treatment plant. Treatment of Sewage: Screening, Types of screens, Grit removal, Skimming, Sedimentation of sewage, Aerobic and anaerobic process, Sludge digestion, trickling filters, Activated sludge process, Disposal of sewage, Oxidation Pond, Oxidation ditch, aerated lagoons, anaerobic lagoon. Rural sanitation: Low cost rural sanitation systems-Privies of a conservancy system of sanitation- Bore hole privy, Aqua privy, Sulabh Shauchalya, Two pit pour flush water seal, Septic tank and soak pit, Recycling and Reuse of domestic waste. Composting (Nadep or Vermiculture), Biogas plants. Low cost community and individual Garbage disposal systems, Ferro-cement storage tanks. Rural sanitation programmes by Government of India, Swachh Bharat Mission-Objectives and strategies. Suchitwa mission. Text / Reference: T/R Book Title/Author S.C. Rangwala: Water Supply and Sanitary Engineering; Charotar Publishing T1

House R1 Garg, S.K., Environmental Engineering Vol. I and Vol. II, Khanna Publishers
R2 K.N.Duggal : Elements of Public Health Engineering ; S.Chand & Co. R3 Sharma
S.C, Environmental Engineering, Khanna Publishing House, New Delhi2 R4 Birdie, G.
S. and Birdie, J.S.Water Supply and Sanitary Engineering, Dhanpat Rai R5 Desai,
Vasant, Rural Development in India: Past, Present and Future: a challenge in the
crisis, Himalaya Publishing House, Delhi. R6 Singh, Katar, Rural Development
Principles, Policies and Management, Sage Publications India Pvt Ltd. Online

Resources: Sl.No Website Link 1 <http://nptel.ac.in/courses> 2

<http://www.vlab.co.in/ba-nptel-labs-civil-engineering> 3 Document Published by
Ministry of Rural development, Govt. of India, Ministry of Rural development.

परिभाषा/अर्थ definition/meaning परिभाषा/अर्थ. परिभाषा परिभाषा परिभाषा,
steps, application परिभाषा परिभाषा परिभाषा. Technical terms English- परिभाषा
परिभाषा/अर्थ.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Suggest appropriate sewage treatment and disposal methods

	Identify characteristics of sewage	
M4.01		3
Understanding		
	Outline the functions and working of screens,	
M4.02		3
Understanding		
	skimming tanks, and grit chambers.	
	Describe various sewage treatment and disposal	
M4.03		4
Understanding		
	methods	
	Explain various low cost rural sanitation	
M4.04		5
Understanding		
	systems	
	Series Test - II	1

Contents:

Analysis of sewage: Characteristics of sewage, B.O.D., C.O.D. and its significance, Central Pollution Control Board Norms for discharge of treated sewage, Objects of sewage treatment and flow diagram of conventional sewage treatment plant.

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 3 Understanding sources of water. Determine water demands for a water supply. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding sources of water. Determine water demands for a water supply*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Applying scheme Outline standard tests for analyzing water for. (3 marks)

Answer: Begin with the standard definition or identity of *4 Applying scheme Outline standard tests for analyzing water for*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Understanding drinking purposes Explain the different rural water. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding drinking purposes Explain the different rural water*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding supply schemes by the Government Sources of Water: Surface source - lakes, streams, rivers and impounded reservoirs, Yield from surface source - Underground sources - springs, wells, infiltration wells and galleries - Yield from wells - test for yield. Intake Structures, Definition and types, Factors governing the location of an intake structure, Types of intakes. Demand of water: Factors affecting rate of demand, Variations of water demands, Forecasting of population, Methods of forecasting of population, (Simple problems on forecasting of population), Design period, Estimating of quantity of water supply required for city or town. Quality of water: Need for analysis of water, Characteristics of water - Physical, Chemical and Biological, Testing of water for Total solids, hardness, chlorides, dissolved Oxygen, pH, Fluoride, Nitrogen and its compounds,

Bacteriological tests, E coli, B coli index, MPN, Sampling of water, Water quality standards as per IS 10500. Water supply schemes - Objectives, components, Rural water supply schemes by Government-NRDWP-Jal Jeevan Mission, Atal Bhujal Yojana, Jal Sakthi Abhiyan KRWSA-Jalanidhi, Jala Subhiksha.. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding supply schemes by the Government Sources of Water: Surface source - lakes, streams, rivers and impounded reservoirs, Yield from surface source - Underground sources - springs, wells, infiltration wells and galleries - Yield from wells - test for yield. Intake Structures, Definition and types, Factors governing the location of an intake structure, Types of intakes. Demand of water: Factors affecting rate of demand, Variations of water demands, Forecasting of population, Methods of forecasting of population, (Simple problems on forecasting of population), Design period, Estimating of quantity of water supply required for city or town. Quality of water: Need for analysis of water, Characteristics of water - Physical, Chemical and Biological, Testing of water for Total solids, hardness, chlorides, dissolved Oxygen, pH, Fluoride, Nitrogen and its compounds, Bacteriological tests, E coli, B coli index, MPN, Sampling of water, Water quality standards as per IS 10500. Water supply schemes - Objectives, components, Rural water supply schemes by Government-NRDWP-Jal Jeevan Mission, Atal Bhujal Yojana, Jal Sakthi Abhiyan KRWSA-Jalanidhi, Jala Subhiksha..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 2

Define or explain sedimentation, and sedimentation with 3 Understanding coagulation. Outline the construction and operation of slow. (3 marks)

Answer: Begin with the standard definition or identity of *sedimentation, and sedimentation with 3 Understanding coagulation. Outline the construction and operation of slow.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 3 Understanding sand, rapid sand and pressure filters. Explain methods of disinfection and. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding sand, rapid sand and pressure filters. Explain methods of disinfection and.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain advanced treatment methods for the water 3 Understanding in a given situation Identify the different components of a. (3 marks)

Answer: Begin with the standard definition or identity of *advanced treatment methods for the water 3 Understanding in a given situation Identify the different components of a.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 3 Understanding conveyance system Explain with sketches the methods of Applying. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding conveyance system Explain with sketches the methods of Applying.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain 3 Remembering sullage, and garbage. (3 marks)

Answer: Begin with the standard definition or identity of *3 Remembering sullage, and garbage*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Identify the different building sanitary fittings 4 Understanding and systems of plumbing Explain different sewerage. (3 marks)

Answer: Begin with the standard definition or identity of *Identify the different building sanitary fittings 4 Understanding and systems of plumbing Explain different sewerage*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain systems. (3 marks)

Answer: Begin with the standard definition or identity of *systems*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Describe the various types of sewer 3 Understanding appurtenances Building Sanitation: Necessity of sanitation, Necessity to treat domestic sewage, Definitions - Sewage, sullage, types of sewage. Definition of the terms related to Building Sanitation- Water pipe, Rain water pipe, Soil pipe, Sullage pipe, Vent pipe. Building Sanitary fittings- Water closet - Indian and European type, flushing cistern, wash basin, sinks, Urinals. Traps-types, qualities of good trap. Systems of plumbing - one pipe, two pipe, single stack, choice of system. Principles regarding design of building drainage, inspection and junction chambers, their necessity, location, size and shape. Systems of Sewerage and Sewer Appurtenances: Types of Sewers, Systems of sewerage, self-cleansing velocity and non-scouring velocity, Laying, Testing and maintenance of sewers, Manholes and Drop Manhole-component parts, location, spacing, construction details, Sewer Inlets, Street Inlets.. (3 marks)

Answer: Begin with the standard definition or identity of *Describe the various types of sewer 3 Understanding appurtenances Building Sanitation: Necessity of sanitation, Necessity to treat domestic sewage, Definitions - Sewage, sullage, types of sewage. Definition of the terms related to Building Sanitation- Water pipe, Rain water pipe, Soil pipe, Sullage pipe, Vent pipe. Building Sanitary fittings- Water closet - Indian and European type, flushing cistern, wash basin, sinks, Urinals. Traps-types, qualities of good trap. Systems of plumbing - one pipe, two pipe, single stack, choice of system. Principles regarding design of building drainage, inspection and junction chambers, their necessity, location, size and shape. Systems of Sewerage and Sewer Appurtenances: Types of Sewers, Systems of sewerage, self-cleansing velocity and non-scouring velocity, Laying, Testing and maintenance of sewers, Manholes and Drop Manhole-component parts, location, spacing, construction details, Sewer Inlets, Street Inlets..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 4

Define or explain 3 Understanding Outline the functions and working of screens,. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding Outline the functions and working of screens,.* State the governing principle, list the key

components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding skimming tanks, and grit chambers. Describe various sewage treatment and disposal. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding skimming tanks, and grit chambers. Describe various sewage treatment and disposal*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Understanding methods Explain various low cost rural sanitation. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding methods Explain various low cost rural sanitation*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 5 Understanding systems Analysis of sewage: Characteristics of sewage, B.O.D., C.O.D. and its significance, Central Pollution Control Board Norms for discharge of treated sewage, Objects of sewage treatment and flow diagram of conventional sewage treatment plant. Treatment of Sewage: Screening, Types of screens, Grit removal, Skimming, Sedimentation of sewage, Aerobic and anaerobic process, Sludge digestion, trickling filters, Activated sludge process, Disposal of sewage, Oxidation Pond, Oxidation ditch, aerated lagoons, anaerobic lagoon. Rural sanitation: Low cost rural sanitation systems-Privies of a conservancy system of

sanitation- Bore hole privy, Aqua privy, Sulabh Shauchalya, Two pit pour flush water seal, Septic tank and soak pit, Recycling and Reuse of domestic waste. Composting (Nadep or Vermiculture), Biogas plants. Low cost community and individual Garbage disposal systems, Ferro-cement storage tanks. Rural sanitation programmes by Government of India, Swachh Bharat Mission- Objectives and strategies. Suchitwa mission. Text / Reference: T/R Book Title/Author S.C. Rangwala: Water Supply and Sanitary Engineering; Charotar Publishing T1 House R1 Garg, S.K., Environmental Engineering Vol. I and Vol. II, Khanna Publishers R2 K.N.Duggal : Elements of Public Health Engineering ; S.Chand & Co. R3 Sharma S.C, Environmental Engineering, Khanna Publishing House, New Delhi2 R4 Birdie, G. S. and Birdie, J.S.Water Supply and Sanitary Engineering, Dhanpat Rai R5 Desai, Vasant, Rural Development in India: Past, Present and Future: a challenge in the crisis, Himalaya Publishing House, Delhi. R6 Singh, Katar, Rural Development Principles, Policies and Management, Sage Publications India Pvt Ltd. Online Resources: Sl.No Website Link 1 <http://nptel.ac.in/courses> 2 <http://www.vlab.co.in/ba-nptel-labs-civil-engineering> 3 Document Published by Ministry of Rural development, Govt. of India, Ministry of Rural development.. (3 marks)

Answer: Begin with the standard definition or identity of 5 Understanding systems Analysis of sewage: Characteristics of sewage, B.O.D., C.O.D. and its significance, Central Pollution Control Board Norms for discharge of treated sewage, Objects of sewage treatment and flow diagram of conventional sewage treatment plant. Treatment of Sewage: Screening, Types of screens, Grit removal, Skimming, Sedimentation of sewage, Aerobic and anaerobic process, Sludge digestion, trickling filters, Activated sludge process, Disposal of sewage, Oxidation Pond, Oxidation ditch, aerated lagoons, anaerobic lagoon. Rural sanitation: Low cost rural sanitation systems-Privies of a conservancy system of sanitation- Bore hole privy, Aqua privy, Sulabh Shauchalya, Two pit pour flush water seal, Septic tank and soak pit, Recycling and Reuse of domestic waste. Composting (Nadep or Vermiculture), Biogas plants. Low cost community and individual Garbage disposal systems, Ferro-cement storage tanks. Rural sanitation programmes by Government of India, Swachh Bharat Mission-Objectives and strategies. Suchitwa mission. Text / Reference: T/R Book Title/Author S.C. Rangwala: Water Supply and Sanitary Engineering; Charotar Publishing T1 House R1 Garg, S.K., Environmental Engineering Vol. I and Vol. II, Khanna Publishers R2 K.N.Duggal : Elements of Public Health Engineering ; S.Chand & Co. R3 Sharma S.C, Environmental Engineering, Khanna Publishing House, New Delhi2 R4 Birdie, G. S. and Birdie, J.S.Water Supply and Sanitary Engineering, Dhanpat Rai R5 Desai, Vasant, Rural Development in India: Past, Present and Future: a challenge in the crisis, Himalaya Publishing House, Delhi. R6 Singh, Katar, Rural Development Principles, Policies and Management, Sage Publications India Pvt Ltd. Online Resources: Sl.No Website Link 1 <http://nptel.ac.in/courses> 2 <http://www.vlab.co.in/ba-nptel-labs-civil-engineering> 3 Document Published by Ministry of Rural development, Govt. of India, Ministry of Rural development.. State the

governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **3 Understanding sources of water. Determine water demands for a water supply.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **sedimentation, and sedimentation with 3 Understanding coagulation. Outline the construction and operation of slow.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **3 Remembering sullage, and garbage.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **3 Understanding Outline the functions and working of screens,.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link <http://nptel.ac.in/courses>
- <http://www.vlab.co.in/ba-nptel-labs-civil-engineering>
- Document Published by Ministry of Rural development, Govt. of India, Ministry
- of Rural development.

IN-PAGE SEARCH

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